

Fairness-Aware Routing in Quantum Networks & Clinical Settings

via Multi-Armed, Contextual, and Informative Bandit Methods

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Abstract—Sequential decision systems increasingly determine how scarce diagnostic, computational, and communication resources are routed under uncertainty. This project studies fairness-aware contextual multi-armed bandits for such decisions when context is missing, delayed, noisy, or group-dependent. The central claim is that clinical diagnostic decision workflows and quantum network routing share the same structural failure mode: policies must choose an action from partial feedback, yet the groups or flows with the least reliable context can absorb the worst errors. The work develops a reusable evaluation framework for non-contextual bandits, contextual bandits, and informative contextual bandits across two executable environment families. The quantum path reuses the legacy quantum-routing evaluation stack and saved-state paths from the quantum-routing evaluation paper, while the clinical path now runs through an embedded synthetic clinical environment with clinical models, resource allocation, EQUITAS mediation, and governed fairness artifacts. Work completed so far includes the related-work synthesis, domain model, architecture specification, embedded clinical evaluator path, legacy quantum fairness sidecar, default and fairness reporting datasets, validation-hub artifacts, and CI-protected smoke gates. The contribution is a reproducible method for measuring and mitigating time-evolving fairness disparities while preserving utility under non-stationarity and limited context.

I. INTRODUCTION

Many applied data science systems are not one-shot predictors. They are sequential decision systems that route limited resources under uncertainty: which diagnostic test to order, which case to escalate, which model or pipeline to trust, or which quantum-network path should receive scarce entanglement-generation attempts. These decisions are difficult because the system observes only partial feedback. The chosen action reveals an outcome, but the unchosen alternatives remain counterfactual. They are also fairness critical because the quality of the observed context is often uneven across groups. When one population has delayed testing, incomplete patient history, lower-quality measurements, or less representative calibration data, a policy optimized only for aggregate reward can hide subgroup errors and compound disparities over time.

This project frames that problem as fairness-aware contextual bandit learning. At each round, a policy observes context, chooses an action, receives reward only for the chosen action, and updates its future decisions. The scientific question is not only whether contextual information improves reward, but whether informative context and fairness-aware monitoring reduce disparities when context is missing or degraded. This

matters in clinical workflows because false-negative or delayed-escalation gaps can produce direct harm. It also matters in quantum-network routing because future communication systems will allocate scarce probabilistic resources among competing flows; fairness in success probability and latency is a service-quality issue rather than a purely normative add-on.

The study develops a shared environment–runner–evaluator architecture for two complementary routing families. The first is a quantum-routing testbed based on hybrid contextual bandits for entanglement routing and qubit allocation [1]. It exposes noisy links, uncertain entanglement generation, adversarial or non-stationary conditions, allocator decisions, and saved evaluator state. The second is an embedded clinical simulation motivated by equitable bioinformatics and fairness auditing work [2]. It models test ordering, retesting, escalation, diagnostic attention, model-assisted triage, site-level constraints, and cohort-level context quality. The shared abstraction is that each domain routes scarce resources from partial feedback while fairness risk emerges from unequal context quality.

Four research questions drive the work. RQ1 asks how much richer context improves utility and fairness relative to non-contextual baselines. RQ2 asks how quickly disparities emerge when a group, site, cohort, or flow class receives noisier, delayed, or incomplete context. RQ3 asks whether missingness-aware or fairness-mediated policy updates reduce outcome gaps without unacceptable utility loss. RQ4 asks whether the same fairness mediation strategy transfers between clinical diagnostic simulation and quantum routing, or whether mitigation must remain domain-specific. Preliminary evidence supports the context-gain premise: predictive-context quantum policies improve over no-context baselines by 3.74 percentage points in the internal corpus and 2.83 percentage points in the standardized external corpus, while the clinical EQUITAS evidence package shows large improvements after domain context restoration and under-represented-context augmentation.

II. RELATED WORK

A. Bandits and Contextual Sequential Decision-Making

Multi-armed bandits model online learning under partial feedback, balancing exploration and exploitation while observing rewards only for selected actions. UCB-style algorithms provide finite-time regret guarantees in stochastic settings [3], while Thompson sampling offers a Bayesian alternative that often

performs well empirically [4]. The broader bandit literature provides the regret and uncertainty-estimation foundation for this project [5]. Contextual bandits extend this model by conditioning action choice on observed side information. LinUCB and related methods demonstrate how context can improve online personalization and decision quality [6], [7], and reduction-based contextual bandit approaches show how richer policy classes can be scaled [8]. However, most of this work optimizes expected reward and assumes that context is observed correctly or representatively.

Offline policy evaluation is also relevant because the final framework will compare policies from logged simulator outputs as well as from fresh online runs. Doubly robust estimators combine reward modeling with inverse-propensity weighting and can reduce variance when evaluating bandit policies from logged data [9]. In this project, those ideas motivate careful logging of actions, propensities or selection rules, context snapshots, and oracle comparisons. They also highlight a fairness concern: if one group has less support in the logged data, group-specific estimates can be noisier even when the aggregate estimator appears stable.

B. Fairness, Missing Context, and Non-Stationarity

Fairness in supervised learning is often expressed through group-conditioned error criteria such as equality of opportunity and equalized odds [10]. In sequential settings, the problem is harder: fairness constraints affect exploration, data collection, and the future evidence available to the learner. Joseph et al. connect fairness to bandit learning directly [11], while Chen et al. study fair contextual multi-armed bandits with explicit group constraints [12]. Offline contextual-bandit fairness work shows how importance-weighted estimators can support high-probability constraints [13], but online distribution shift and time-evolving disparity remain open practical challenges. Non-stationary bandit work addresses changing reward distributions through discounting, change detection, or variation budgets [14], [15]. Limited or unobserved context work similarly motivates algorithms that remain robust when features are unavailable or latent [16], [17].

The limitation across these areas is that they often address only one axis of the final problem. Standard bandits provide regret machinery but not fairness audits. Fairness definitions clarify what disparities matter but are often studied in static supervised settings. Non-stationary bandits handle changing reward processes but usually report aggregate utility. Restricted-context work handles missing features but rarely treats missingness as a group-dependent fairness mechanism. The present project combines these axes by treating missing context, time-varying reward, and group disparity as simultaneous properties of the same learning process.

C. Clinical Fairness and Quantum Routing

Clinical algorithmic bias studies show that proxy targets and measurement processes can encode structural inequities, producing under-allocation for some groups even when aggregate performance appears strong [18]. This motivates outcome-

TABLE I
HOW THE REVIEWED LITERATURE SUPPORTS THE PROJECT.

Literature area	Role in this study
Bandit foundations	partial feedback, regret & baselines
Contextual bandits	model side data & missing-context sensitivity
Fairness definitions	outcome gaps, FNR/FPR & service equity
Non-stationarity	shift scenarios & time-based disparity curves
Clinical bias studies	subgroup error monitor & proxy-risk caution
Quantum routing	resource-scarce routing testbed & stateful simulator

based disparity monitoring such as false-negative and false-positive gaps. In parallel, quantum-network research describes an emerging communication infrastructure in which routing decisions are governed by probabilistic entanglement generation, fidelity, storage limits, and scarce resources [19]–[21]. Prior quantum routing work usually emphasizes throughput, fidelity, or robustness rather than group-based service equity. The gap this project addresses is therefore the intersection of contextual bandit learning, missing context, non-stationarity, and explicit fairness monitoring across both clinical and quantum-resource routing domains.

Table I summarizes the role of each category. The final analysis will expand this discussion with direct comparisons once the experiments are complete, but the literature already establishes the scientific gap: existing work rarely evaluates outcome-based fairness under missing context and distribution shift across two structurally similar but domain-distinct routing systems.

III. APPROACH AND METHODOLOGY

A. Shared Missing-Context Routing Model

The methodology treats both domains as routing under missing context. Business Understanding identifies the shared harm mechanism: clinical and quantum systems route scarce resources under uncertainty, and unequal context quality can produce repeated allocation harm. Data Understanding identifies the measurable patterns that make that harm testable, including group, site, cohort, service line, test distribution, escalation action, latency, reward, success, scenario, allocator, route choice, and evaluator-state fields.

In the clinical domain, the routed resource is diagnostic attention: scarce tests, retests, triage escalation, model-assisted prioritization, or model/pipeline selection. In the quantum domain, the routed resource is entanglement-generation effort, qubit allocation, or path capacity. In both domains, the policy observes partial context, chooses an action, receives partial feedback, and updates. The key experimental variable is the richness and reliability of context available to the policy.

Figure 1 summarizes the shared model. Phase 1 defines the environment and available resources. Phase 2 determines what context is observed and which context is missing or degraded. Phase 3 selects an action using a non-contextual, contextual, or informative contextual bandit policy. Phase 4 evaluates both performance and fairness, then feeds mitigation

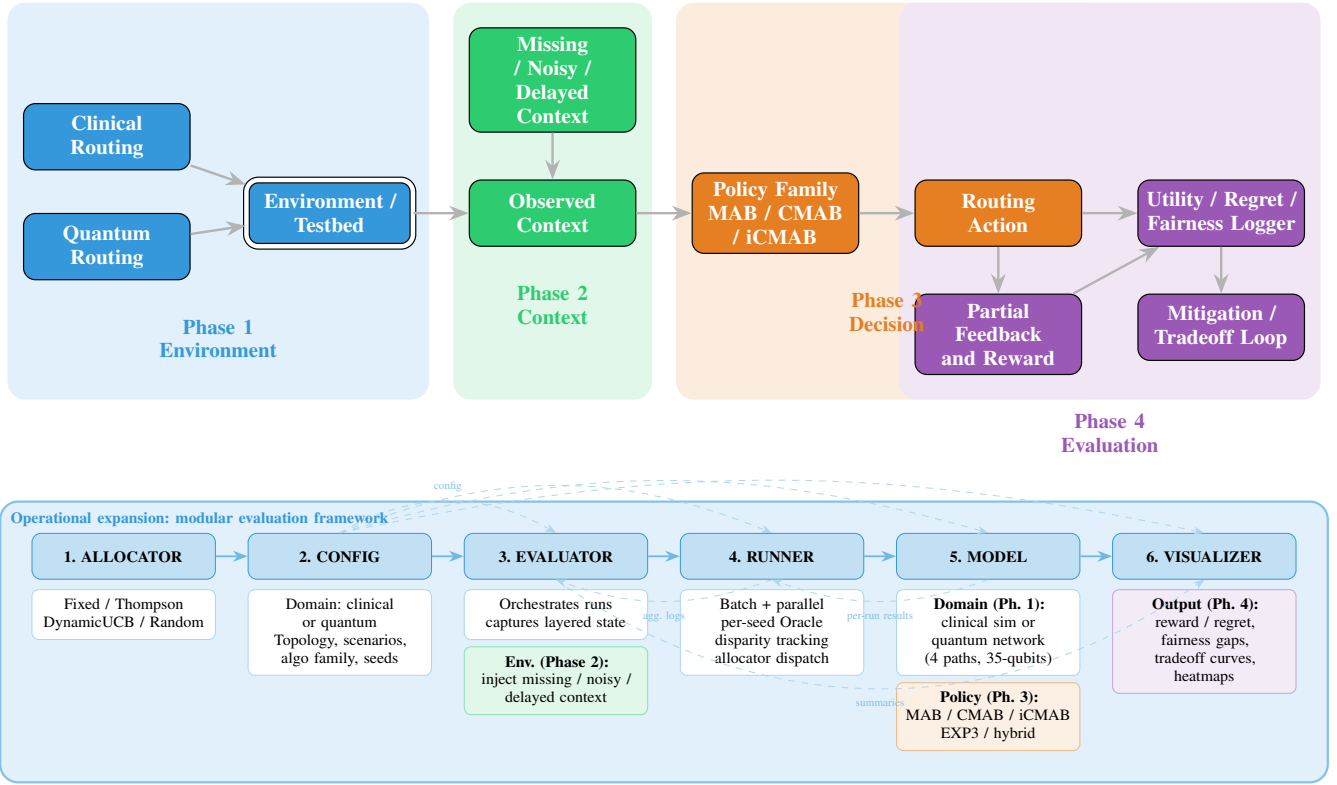


Fig. 1. Shared domain model and operational evaluation framework. The top panel isolates the causal chain that can generate unfair outcomes in both domains: the environment defines the routing problem, missing/noisy/delayed information perturbs only the context stage, the policy family acts on the degraded observation, and utility–fairness logging plus mitigation occur only after partial feedback is received. Clinical actions map to test ordering, retesting, or escalation; quantum actions map to route or allocator decisions. The bottom panel expands the implementation stack and highlights the three experimental intervention points used throughout the project: context degradation in the evaluator environment, policy-family choice in the model layer, and utility–fairness tradeoff analysis in the visualizer.

or updated policy information back into future decisions. This model is useful because it makes the failure point explicit: if context is systematically lower-quality for one group or flow class, the learning process can convert measurement inequity into repeated allocation inequity. It also clarifies where the experiments intervene: missingness is injected before the policy sees the state, while mitigation is evaluated only after utility and disparity have been logged.

At round t , the learner receives an observed context vector $\tilde{\mathbf{x}}_t$, which may differ from the latent complete context $\mathbf{x}_t$ because of missingness, delay, or measurement noise. It chooses an action a_t and observes reward $r_t(a_t)$ only for that action. Utility is tracked through cumulative regret,

$$R_T = \sum_{t=1}^T (r_t(a_t^*) - r_t(a_t)), \quad (1)$$

where a_t^* is the best available action under the simulator oracle. Fairness is tracked through time-indexed disparity gaps,

$$\Delta_m(t) = |m_t(G=0) - m_t(G=1)|, \quad (2)$$

where m_t is a domain-specific outcome metric such as false-negative rate, success probability, latency, or escalation delay. This notation separates the learning objective from the fairness

audit: a policy can have low regret in aggregate while still producing high $\Delta_m(t)$ for specific groups.

B. Bandit Policies and EQUITAS Mediation

The baseline non-contextual policy is an upper-confidence-bound bandit. For action a , let $N_{a,t}$ be the number of times action a has been selected before round t and let $\hat{\mu}_{a,t}$ be its empirical reward mean. The UCB score is

$$U_t^{\text{MAB}}(a) = \hat{\mu}_{a,t-1} + \alpha \sqrt{\frac{2 \log t}{\max(1, N_{a,t-1})}}, \quad (3)$$

and the learner selects $a_t = \arg \max_a U_t^{\text{MAB}}(a)$. This baseline deliberately ignores patient, site, topology, load, or fidelity features, so it measures the cost of acting without context.

The contextual policy replaces scalar action means with action-specific linear models. For each action, let $A_{a,t}$ and $b_{a,t}$ denote the accumulated feature covariance and reward-weighted feature vector, with $\hat{\theta}_{a,t} = A_{a,t}^{-1} b_{a,t}$. Given observed context $\tilde{\mathbf{x}}_t$, the contextual UCB score is

$$U_t^{\text{CMAB}}(a) = \tilde{\mathbf{x}}_t^\top \hat{\theta}_{a,t-1} + \alpha \sqrt{\tilde{\mathbf{x}}_t^\top A_{a,t-1}^{-1} \tilde{\mathbf{x}}_t}. \quad (4)$$

After observing reward $r_t(a_t)$, the selected action updates as $A_{a_t,t} = A_{a_t,t-1} + \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t^\top$ and $b_{a_t,t} = b_{a_t,t-1} + r_t(a_t) \tilde{\mathbf{x}}_t$.

EQUITAS adds fairness mediation as a governed scoring layer around either bandit family. Let $U_t(a)$ be the utility score from Eq. 3 or Eq. 4, $\hat{\Delta}_{m,t}(a)$ be the estimated disparity gap that would result from action a , ϵ be an allowable gap, and $C_t(a)$ be a context-deficit penalty for missing, stale, or low-confidence features. The mediated score is

$$S_t^{\text{EQUITAS}}(a) = U_t(a) - \lambda \max(0, \hat{\Delta}_{m,t}(a) - \epsilon) - \gamma C_t(a), \quad (5)$$

with action $a_t = \arg \max_a S_t^{\text{EQUITAS}}(a)$. The parameters λ and γ control the utility–fairness and utility–context–quality tradeoffs. In clinical routing, $\hat{\Delta}$ tracks error, escalation, or delay gaps across sites and cohorts. In quantum routing, it tracks service gaps such as success probability, latency, or resource access across flows or service classes.

C. Design Decisions

Bandits are used instead of full reinforcement learning because both testbeds provide bandit feedback: only the chosen diagnostic or routing action reveals an outcome. The project compares three policy families. Non-contextual MABs serve as a baseline for action-only learning. Contextual bandits test whether observed context improves utility and equity. Informative contextual bandits represent policies that recover, augment, or better weight missing context. The simulation-first clinical design avoids sensitive patient data while allowing controlled missingness and group-specific measurement noise. The quantum testbed reuses an existing, domain-grounded routing environment so that fairness analysis is built on top of realistic uncertainty rather than a toy network.

Fairness is measured as an online outcome property. In diagnostics, the target metrics are group-wise false-negative and false-positive gaps, along with utility and cost-sensitive reward. In quantum routing, service equity is measured through success-probability and latency gaps across flow groups. Mitigation mechanisms under consideration include fairness-regularized policy updates, group-aware calibration or thresholding, and missingness-aware context augmentation. These mechanisms will be evaluated as tradeoffs rather than as free improvements, because fairness constraints can reduce short-term reward while improving reliability and equity.

D. Planned Mitigation Strategy

The first mitigation family will treat fairness as a penalty or constraint in the policy-update loop. In the simplest form, the learner receives a utility reward and a disparity signal; policy updates are then adjusted when recent group-specific outcomes exceed an allowed gap. This makes fairness operational during learning instead of applying it only after the experiment. A second family will use missingness-aware context augmentation. If a group or flow class systematically receives incomplete context, the policy can either acquire additional features, impute missing indicators, or adjust confidence estimates so that uncertainty caused by measurement gaps is not mistaken for low expected value. A third family will test group-aware calibration or thresholding at the decision boundary. This is especially

relevant for diagnostic-like escalation, where a small threshold shift can change false-negative rates.

The mitigation comparison will be conservative. The analysis will not claim that lower disparity is automatically better if the policy produces unsafe or unusable utility loss. Instead, each mitigation will be evaluated through utility–fairness tradeoff curves. The preferred policy is expected to be one that preserves most of the utility of contextual learning while reducing time-evolving disparity under missing context. This is why the project compares MAB, CMAB, and iCMAB policies under the same seeds, scenario settings, and logged artifacts: the point is to isolate the effect of context quality and fairness-aware updates rather than confounding algorithm choice with simulator configuration.

IV. IMPLEMENTATION AND REPRODUCIBILITY

A. Architecture

The implementation realizes the method in Table II through a modular experiment architecture. Each environment supplies domain state, available actions, outcome semantics, and context-quality indicators. A runner executes one or more policy models inside that environment. A resource allocator controls scarce capacity before actions are selected, such as qubit/path capacity in quantum routing or test/retest/escalation capacity in the clinical setting. An evaluator coordinates repeated runs, collects states, and computes utility, regret, and disparity summaries. EQUITAS sits between policy scoring and evaluation: it can mediate action scores during a run, or it can audit saved states after a legacy run without changing the original execution semantics.

Table IV summarizes the core components. The design is intentionally object-oriented because clinical and quantum experiments share a control pattern but not the same domain semantics. The quantum environment defines network topologies, link uncertainty, path decisions, allocator settings, and routing outcomes. The clinical environment defines sites, cohorts, service lines, diagnostic actions, resource constraints, and diagnostic outcomes. Both environments emit comparable state records so the same evaluator and fairness reporting strategy can operate across domains.

B. State Persistence and Reporting

The system treats state as a first-class research artifact. Model state captures policy-level information needed to resume or inspect a learner. Runner state captures the sequence of environment interactions for a specific configuration, seed, and allocator setting. Evaluator state captures cross-run summaries needed for comparing policies and scenarios. Reporting state captures the derived datasets, fairness summaries, visualizations, and manifests needed to reproduce claims. These layers prevent a long-running experiment from becoming a single opaque metric and allow fairness analyses to be repeated as definitions or mitigation strategies evolve.

TABLE II
EQUITAS-MEDIATED FAIRNESS-AWARE ROUTING PROCEDURE.

Input: environment $\mathcal{E}$, action set $\mathcal{A}$, policy family π , fairness metric m , allowable gap ϵ , tradeoff weights λ, γ , horizon T .

1. Initialize per-action bandit statistics, fairness summaries, context-quality summaries, and state provenance.
2. **For** $t = 1, \dots, T$: observe incomplete context $\tilde{\mathbf{x}}_t$ and context-quality indicators from $\mathcal{E}$.
3. Compute a utility score $U_t(a)$ for each $a \in \mathcal{A}$ using the MAB score in Eq. 3 or the contextual score in Eq. 4.
4. Estimate the group- or flow-conditioned disparity effect $\hat{\Delta}_{m,t}(a)$ and context-deficit penalty $C_t(a)$.
5. Apply EQUITAS mediation with Eq. 5; choose $a_t = \arg \max_a S_t^{\text{EQUITAS}}(a)$.
6. Execute a_t , observe partial reward $r_t(a_t)$ and outcome fields, then update policy statistics only for the selected action.
7. Persist the round state: context, action, reward, outcome, group/flow labels, allocator metadata, EQUITAS score components, and provenance.
8. **Return:** default utility summaries, fairness summaries, state artifacts, and reporting datasets for clinical or quantum analysis.

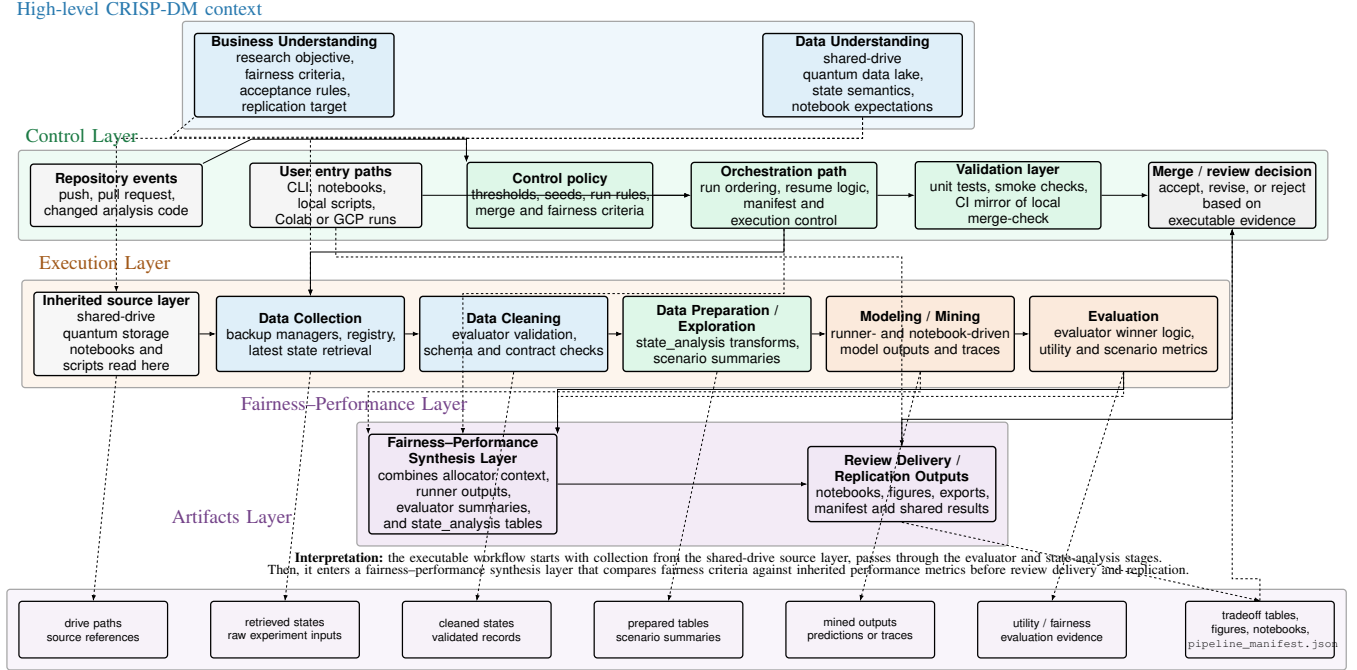


Fig. 2. Architecture diagram adapted from the architecture report. The workflow is an evidence-flow view: domain context enters through an environment abstraction, runner/model/allocator decisions produce stateful outcomes, the EQUITAS layer converts those outcomes into utility and fairness evidence, and the reporting layer emits datasets, figures, manifests, and reproducibility artifacts. The same flow supports clinical simulation and quantum routing without forcing either domain to imitate the other.

C. Validation Strategy

Validation is organized around contracts rather than ad hoc scripts. Stage tests verify that data collection, cleaning, preparation, modeling, evaluation, and reporting artifacts satisfy their schemas. Experiment tests verify that clinical and quantum runs emit non-empty state records with the fields required for utility and fairness analysis. Sidecar tests verify that legacy quantum states can be converted into canonical fairness outcomes without changing old execution behavior. Reporting tests verify that master datasets, figures, and manifests remain linked to their source states. This strategy connects software correctness to scientific reproducibility: the same state contracts that protect code changes also preserve the evidence needed to interpret preliminary findings.

V. DATA SETS AND EXPERIMENTAL INPUTS

The current quantum testbed uses simulated quantum-network routing scenarios with probabilistic link success, qubit-capacity constraints, allocator settings, time scales, and stochastic or adversarial conditions. The key outputs are framework-state and model-state artifacts, reward traces, oracle comparisons, winner summaries, efficiency measures, and scenario-level statistics. These artifacts are stored locally or in a shared Drive data lake depending on the execution path.

Table V summarizes the main datasets already available in the quantum framework. The project inherits a validated quantum-routing corpus that already contains scenario-level and experiment-level outputs for four external testbeds and for the internal hybrid framework. These corpora are paired with shared data-lake state artifacts for model and framework

TABLE III
APPLICATION MAPPING FOR THE SHARED METHOD.

Element	Clinical setting	Quantum setting
Context	site, cohort, service line, test signals, missingness	topology, load, fidelity, route state, capacity
Action	test, retest, escalation, model-assisted triage	path, route, qubit allocation, retry policy
Reward	diagnostic utility, delay/cost penalty, error outcome	success, latency, fidelity, Oracle-normalized efficiency
Fairness gap	FNR/FPR, escalation delay, access imbalance	success-rate, latency, or service-class imbalance

TABLE IV
HIGH-LEVEL IMPLEMENTATION ARCHITECTURE.

Component	Responsibility
Environment abstraction	exposes domain state, actions, rewards, groups, and context quality
Runner	executes model policies over repeated rounds and seeds
Allocator/resource policy	controls scarce test, routing, qubit, or escalation capacity
EQUITAS mediator	applies fairness and context-quality adjustments to policy evidence
Evaluator	aggregates run states into utility, regret, and disparity evidence
State/reporting layer	persists model, runner, evaluator, fairness, and figure artifacts

persistence. This matters because the fairness-aware analysis can build directly on preserved runner/evaluator artifacts.

A small synthetic validation corpus preserves the shape of the final fairness problem without requiring private data. Each record includes a group label, signal feature, contextual feature, binary outcome, missingness flag, and invalid-record flag. The dataset is deliberately small so controlled validation remains fast, but it exercises the same operations needed by the final project: cleaning missing context, preparing train/test artifacts, producing predictions, computing group metrics, and writing final summaries.

The planned clinical diagnostic testbed will also be simulation-first. Its records will represent patient or case contexts, protected or fairness-relevant groups, diagnostic action choices, observed outcomes, test costs, delay or escalation signals, and injected missingness or measurement noise. This design allows controlled evaluation of false-negative and false-positive gaps before moving to any approved open or sensitive clinical data.

The clinical side also has current preliminary evidence from the EQUITAS bioinformatics project [2], which is used here as a diagnostic-relevant source package. The source package centers on RNA secondary-structure prediction with fairness auditing over sequence type, length, and GC-content groups. Its current corpus contains 20 RNA sequences after fairness-

aware augmentation, including tRNA, rRNA, miRNA, synthetic controls, and 10 targeted Synthetic LowGC sequences added to repair under-representation. These assets provide an existing clinical-context example for method validation while the final sequential diagnostic testbed remains simulation-first.

Table VI summarizes the common schema. The schema is broad enough to support both testbeds but narrow enough to keep analysis comparable: each record has a decision context, a chosen action, an outcome, and a fairness-relevant group or flow label. This design supports cross-domain transfer analysis because the policy-update and mitigation logic can operate over comparable artifact contracts even when the domain-specific feature semantics differ.

The existing quantum corpora already expose many of the fields the fairness layer needs. The master datasets record allocator, scenario, scenario category, run count, frame configuration, capacity semantics, scale, model identity, reward, regret, Oracle-relative efficiency, retries, failures, misses below threshold, and scenario winners. In practical terms, this means the study has access to real experimental data and preliminary results. The main extension still needed is the fairness layer itself: group or flow labels, subgroup disparity summaries, and time-evolving fairness metrics must be added on top of the already preserved performance corpus.

For the clinical side, the project remains simulation-first. The final clinical dataset will mirror the structure of the quantum corpus: context quality, action choice, binary or graded outcome, cost/delay information, and fairness-relevant group annotation. This keeps the cross-domain comparison meaningful. One dataset is not meant to imitate the domain semantics of the other; rather, both are designed to satisfy the same artifact contract so that the same evaluation and mitigation logic can be applied to both.

VI. ETHICS, FAIRNESS, PRIVACY, AND SECURITY

Ethics is central to the project because the system is explicitly about resource allocation under uncertainty. A policy that optimizes only aggregate reward can still be ethically unacceptable if it shifts errors onto groups that already have lower-quality context. In the clinical setting, the most direct ethical concern is harm from missed or delayed diagnosis. If context is missing because a group has less reliable testing, delayed follow-up, or lower-quality measurements, then a model can learn that the group is harder to classify and route fewer resources toward it. That is not merely a statistical artifact; it is a mechanism by which historical under-measurement becomes future under-allocation.

The project treats fairness as a design requirement rather than as a post-hoc report. Fairness affects how experiments are built, how artifacts are logged, and how results are interpreted. The evaluation must report subgroup metrics, not only average reward or average efficiency. It must preserve context-quality information, because missingness is part of the causal story of disparity. It must also make assumptions visible: which groups are compared, which outcome gaps matter, which simulator mechanisms generate missingness, and

TABLE V
PRIMARY QUANTUM DATASETS AND CORPORA USED BY THE PROJECT.

Corpus	Rows	Cols	Scope	Use in this project
Hybrid master dataset	4,920	30	internal hybrid framework	fairness-layer extension on top of allocator, scenario, reward, regret, and efficiency summaries
Standardized external papers corpus	8,060	32	Papers 2, 7, 8, and 12 at 4K/2K	cross-testbed comparison baseline for adding fairness and transfer analysis
Paper 2 standardized slice	2,450	30	15 nodes, 51 edges, 8 paths	stochastic-noise external benchmark
Paper 7 standardized slice	1,710	30	50 nodes, 141 edges, 15 paths	dense-network external benchmark
Paper 8 standardized slice	1,500	30	20 nodes, 19 edges, 8 paths	compact RL-routing external benchmark
Paper 12 standardized slice	2,400	30	100 nodes, 426 edges, 4 paths	fusion-network external benchmark

TABLE VI
PLANNED DATA FIELDS AND FAIRNESS ROLE.

Field type	Purpose
Group/flow label	subgroup or service class for disparity metrics
Context features	patient/test signals, link/load/fidelity estimates
Missingness flags	degraded, delayed, or unavailable context
Action choice	test, escalation, model, route/allocator decision
Outcome/reward	diagnostic & routing success, latency, or utility
Oracle/baseline	regret, gap & mitigation comparison

TABLE VII
ETHICAL DESIGN REQUIREMENTS AND IMPLEMENTATION RESPONSE.

Requirement	Implementation response
No aggregate-only reporting	utility, subgroup disparity metrics
Expose context quality	log missingness, noise, delay & measurement assumptions
Support accountability	preserve seeds, configs, states, manifests & run provenance
Protect sensitive data	simulation-first data until approved data are available
Run security relevance	quantum routing under unreliable or adversarial conditions

which mitigation thresholds are considered acceptable. Without those details, a reproducible code artifact could still be ethically incomplete because it would hide the value choices that shape the experiment.

Table VII summarizes how ethical concerns translate into design requirements. Privacy and security are secondary analytic goals, but they still shape the project. Clinical data requires privacy-aware handling, which is why the clinical testbed is simulation-first in the current phase. Quantum routing raises a different concern: communication patterns and access to scarce network resources can be sensitive, and adversarial or unreliable conditions can change who receives successful service. The shared framework does not solve every privacy or security problem, but it creates an evaluation structure where those assumptions can be varied and documented rather than ignored.

The EQUITAS preliminary evidence reinforces the same ethical point: under-represented sequence contexts, such as low-

GC examples, required explicit augmentation and re-auditing before fairness claims were defensible. The ethical standard for the final analysis is therefore stronger than “the algorithm works.” The final system should make visible when a policy is accurate but inequitable, robust but unfair to a service class, or fairer only because it sacrifices too much utility to be operationally useful. This tradeoff framing is important for applied data science: fairness is part of the evidence required to justify deployment or to reject a policy as unsafe.

VII. PRELIMINARY RESULTS

A. Quantum Preliminary Results

The preliminary results are organized around the research questions stated in the introduction. The strongest current quantum evidence addresses RQ1: richer context improves routing utility relative to no-context baselines. The quantum side builds on prior threat-aware routing experiments [1], where saved evaluator states, Oracle comparisons, and standardized external corpora already make context-level comparisons possible. These results are not yet full fairness-mitigation results; rather, they identify the utility frontier that EQUITAS must preserve while reducing service disparities.

Figure 3 separates absolute routing quality from the incremental value of added context. The uplift panels show that predictive context is consistently the strongest improvement family, while the size of the gain varies materially by threat regime rather than remaining constant across scenarios.

B. Clinical Preliminary Results

The clinical side contributes preliminary evidence from the EQUITAS bioinformatics source package. This evidence is not presented as final patient-level validation; instead, it supplies a concrete diagnostic-relevant case showing why context availability matters. In this setting, lower context corresponds to structure prediction without thermodynamic/context-aware scoring, while richer context corresponds to Nussinov/MFE feature augmentation, fairness-aware Synthetic LowGC augmentation, and subgroup-aware EQUITAS auditing.

Figure 4 shows that quantum gains are steady and incremental across context levels, whereas the clinical proxy is dominated by a large first-stage recovery when domain-relevant context is restored and then a smaller second-stage gain associated with tuning and fairness-aware augmentation.

TABLE VIII
PRELIMINARY QUANTUM RESULTS AVAILABLE BEFORE FAIRNESS-LAYER EXTENSION.

Evidence source	Current quantitative signal	Why it matters	Role in this study
Threat-aware quantum manuscript	pursuit–neural hybrids outperform non-contextual baselines by roughly 18–24 percentage points in scenario-aggregated efficiency	shows that context-rich routing materially improves utility under realistic threats	establishes the performance baseline to which the fairness layer will be added
Threat-aware quantum manuscript	worst-case performance floors remain above 85% under stochastic threats for the strongest hybrid family	suggests the best policies remain stable in benign-but-noisy settings	provides a natural starting point for testing whether fairness can be improved without breaking robust utility
Threat-aware quantum manuscript	replay-capacity increases can trigger 22–30 percentage-point efficiency collapses under adaptive adversaries	shows that resource policy, not only algorithm choice, shapes deployment quality	motivates fairness-aware evaluation of allocator and capacity decisions, not just bandit model choice
Standardized external corpus	8,060 rows across 4 testbeds, 4 allocators, 5 threat scenarios, 3 scales, and 5 routing models including Oracle	demonstrates breadth of already available quantum evidence	supports cross-testbed fairness benchmarking once subgroup/service labels are defined
Internal hybrid corpus	iCPursuitNeuralUCB currently has the highest mean efficiency in the internal hybrid dataset (86.13%)	identifies the strongest current utility baseline in the internal framework	provides the leading candidate for fairness-aware extension and ablation

TABLE IX
CONTEXT-SPECTRUM SUMMARY FROM THE EXISTING GA CORPORA. GAINS ARE MEASURED AGAINST THE NO-CONTEXT FAMILY WITHIN EACH CORPUS.

Context level	Representative model(s)	Internal mean	Δ vs. no ctx.	External mean	Δ vs. no ctx.
No context	GNeuralUCB, EXPNeuralUCB	82.39%	—	62.83%	—
Context	CPursuitNeuralUCB	84.70%	+2.31 pp	64.11%	+1.28 pp
Predictive context	iCPursuitNeuralUCB	86.13%	+3.74 pp	65.66%	+2.83 pp

TABLE X
CROSS-TESTBED PREDICTIVE-CONTEXT UPLIFT IN THE STANDARDIZED EXTERNAL CORPUS.

Testbed	No ctx.	Context	Pred. ctx.	Pred. gain
Paper 2	72.11	73.18	74.34	+2.23
Paper 7	72.87	73.44	78.39	+5.52
Paper 8	70.05	73.69	73.95	+3.90
Paper 12	41.69	42.23	42.54	+0.85

Table XIII frames the current evidence conservatively. It does not claim that the fairest policy has already been identified. Instead, it shows that the preliminary results are traceable to saved states, fairness artifacts, and reporting manifests. That traceability is essential for RQ2 and RQ3 because time-evolving disparities and mitigation effects cannot be interpreted from aggregate reward alone.

Table VIII makes the quantum evidence concrete. The validated experiment corpus and associated manuscript already provide real model–scenario–allocator–scale evaluations. In the standardized 4K/2K external corpus, the four inherited testbeds contribute 8,060 rows. In the internal hybrid corpus, the strongest current average efficiency belongs to iCPursuitNeuralUCB (86.13%), followed by CPursuitNeuralUCB (84.70%), GNeuralUCB (82.83%), and EXPNeuralUCB (81.95%). These preliminary results identify the current utility frontier and expose which policy

families are worth extending with EQUITAS mediation first.

The context-spectrum view in Table IX and Figure 3 directly addresses RQ1. The no-context condition captures reward-only or non-context baselines, the context condition captures methods that use observed contextual structure, and the predictive-context condition captures informative or forecasting-aware extensions. In both internal and external corpora, performance rises as the method family moves from no context to context to predictive context. The external standardized corpus also shows that gains vary by testbed structure: iCPursuitNeuralUCB reaches approximately 74.34% on Paper 2, 78.39% on Paper 7, 73.95% on Paper 8, and 42.54% on Paper 12. This variation supports the need for cross-testbed fairness analysis rather than a single topology-specific claim.

The clinical evidence addresses the same context hypothesis from a biomedical angle. The EQUITAS package shows that restoring domain-relevant context changes prediction behavior substantially, while under-represented-context augmentation improves parity indicators. These results are preliminary because they come from a clinical proxy rather than the final sequential diagnostic simulator. They are nevertheless methodologically useful: they show that the EQUITAS framing can represent context quality, group representation, utility, and fairness evidence in a form that can later be routed through the same bandit-style evaluator used for quantum experiments.

Together, the preliminary findings support the central novelty of the method. Richer context improves utility, but the

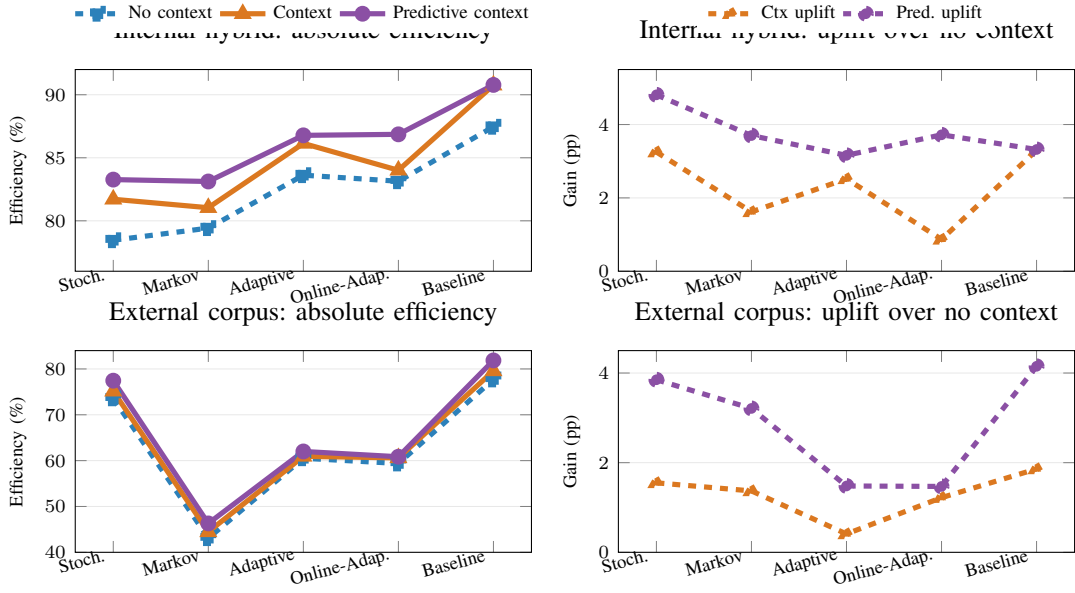


Fig. 3. Quantum context-spectrum results decomposed into absolute performance and marginal uplift. The left column shows the absolute efficiency trajectories for no-context, contextual, and predictive-context families in the internal and external corpora. The right column shows how much contextual and predictive context improve over the corresponding no-context baseline in each scenario. This makes the scientific point explicit: richer context helps everywhere, but the size of the gain is scenario dependent and is usually largest for the predictive-context family.

TABLE XI
PRELIMINARY CLINICAL RESULTS FROM THE EQUITAS BIOINFORMATICS EVIDENCE PACKAGE.

Evidence source	Current quantitative signal	Why it matters	Role in this study
EQUITAS corpus	dataset expanded from 10 to 20 RNA sequences, including 10 Synthetic LowGC sequences	makes clinical-like context gaps measurable in a controlled biomedical proxy task	informs the missingness and group-representation design of the planned simulator
Enhanced Nussinov tuning	Optuna search over 30 trials produced an optimized objective of 56.44 and mean predicted-pair output of 66.07 ± 46.36	shows that richer structural context can materially change diagnostic-relevant prediction behavior	motivates comparing lower-context and richer-context decision policies
Context ablation	thermodynamic/context-aware scoring reported 61.36 ± 44.21 predicted pairs versus 14.96 ± 10.45 without that context	directly supports the context-gain hypothesis in the clinical testbed	anchors the clinical side of the cross-testbed context-gain comparison
Fairness audit	post-augmentation mitigation achieved 6/6 algorithm compliance with the 80% rule, DI ratios of 1.000, and +0.805 average DI improvement	demonstrates that representation/context repair can improve fairness indicators, not only utility	supplies preliminary fairness evidence before final sequential bandit experiments

TABLE XII
AGGREGATE CONTEXT-GAIN COMPARISON ACROSS THE TWO PRELIMINARY EVIDENCE BASES.

Testbed	Lower ctx.	Richer ctx.	Gain
Quantum routing	82.39%	86.13%	+3.74 pp
Clinical bioinformatics	14.96 pairs	66.07 pairs	+51.11 pairs

magnitude of improvement depends on environment structure; fairness-aware mediation is therefore needed to decide whether context gains are distributed equitably. The next empirical step is to apply Eq. 5 during sequential runs and compare utility, regret, and disparity curves against the no-context and

context-only baselines. A mitigation that transfers across both domains would support RQ4; a mitigation that succeeds in one domain and fails in the other would still be scientifically useful because it would identify which parts of fairness mediation are domain-dependent.

VIII. FUTURE WORK

Next, the focus will be on completing the experimental contribution. First, the quantum fairness adapter will connect evaluator and runner outputs to service-equity metrics such as success-probability and latency gaps. This will turn the existing quantum-routing performance infrastructure into a fairness-aware analysis pipeline. Second, the clinical diagnostic simulator will be implemented with controlled missingness, group-dependent measurement noise, diagnostic action choices,

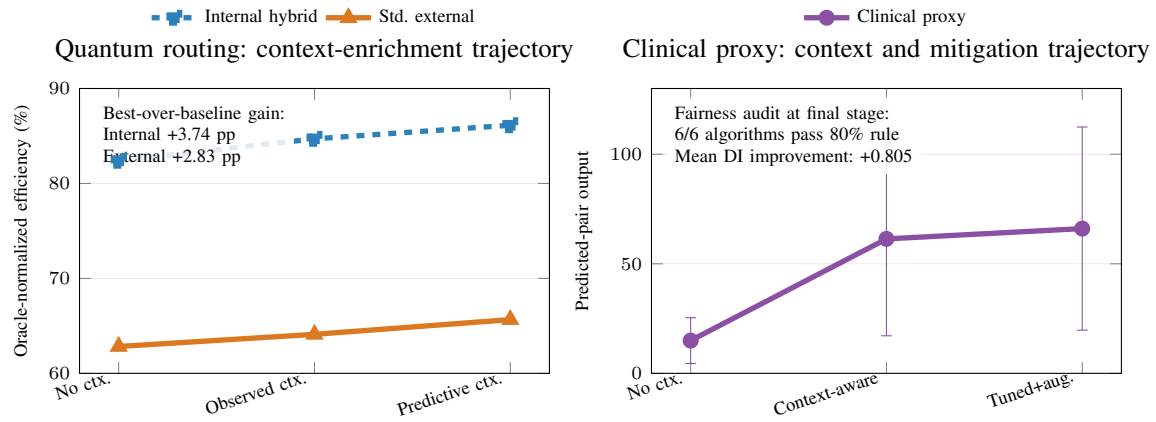


Fig. 4. Cross-testbed context-enrichment trajectories. The left panel shows that quantum-routing performance improves monotonically as the policy moves from no context to observed context to predictive context, with consistent gains in both the internal hybrid corpus and the standardized external corpus. The right panel separates two clinical effects that a single normalized index would obscure: restoring domain-relevant context produces the large primary gain (14.96 to 61.36 predicted pairs), while tuning and fairness-oriented augmentation provide a smaller secondary improvement (66.07 at the final stage) together with improved parity indicators. This decomposition distinguishes pure context recovery from later mitigation or refinement.

TABLE XIII
REPRODUCIBILITY EVIDENCE SUPPORTING PRELIMINARY INTERPRETATION.

Evidence	Interpretation
Quantum state corpora	existing saved runs support context-level utility comparisons
Clinical EQUITAS artifacts	context restoration and augmentation effects are traceable
Fairness sidecar outputs	quantum and clinical states share a governed outcome contract
Reporting manifests	datasets, figures, fairness summaries, and provenance remain linked
Regression tests	state and reporting contracts are protected against implementation drift

and error outcomes. This simulator will allow the project to test false-negative and false-positive gaps without relying on sensitive patient records during development.

Third, at least one mitigation mechanism will be added and compared against utility-only baselines. The initial priority is a simple fairness-regularized or missingness-aware update rule that can be implemented in both domains. Fourth, the same policy stack will be evaluated across both domains to test whether fairness mechanisms transfer. Fifth, the final analysis will add full results, plots, statistical discussion, and a stronger conclusion grounded in experimental evidence. The final analysis will also include a limitations section discussing simulation validity, group-definition choices, and the difference between fairness in a controlled testbed and fairness in deployment.

IX. CONCLUSION

This semester established the foundation for a reproducible fairness-aware contextual bandit project. The work completed so far includes a related-work survey, a shared missing-context domain model, an approach and architecture writeup, a cleaned validation package, embedded clinical and quantum execution

interfaces, and a governed state-reporting layer. The project is positioned to move from design and infrastructure into final experiments. Its expected impact is a practical framework for measuring and mitigating fairness disparities in sequential decision systems where context is incomplete, non-stationary, and unequally distributed across groups.

The remaining work is to replace preliminary evidence with complete sequential experiments. The structure needed for those results is now in place: the literature gap is defined, the domain model is explicit, the architecture is reproducible, the data contracts are identified, and the ethical requirements are stated as part of the technical design.

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